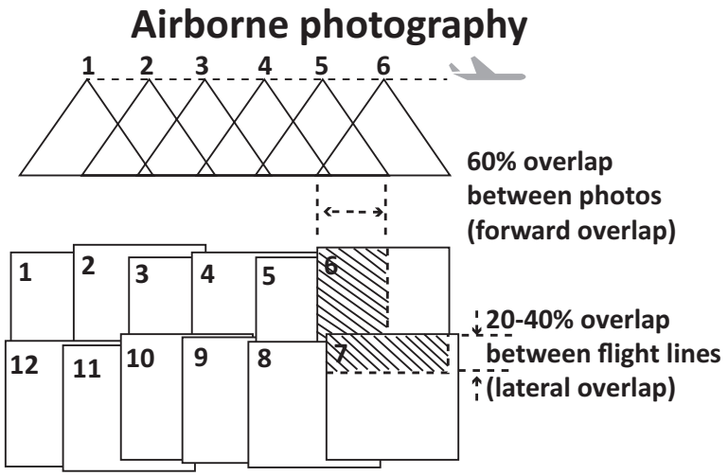



## Components of Remote Sensing

- Energy source
- Transmission path
- Target
- Sensor

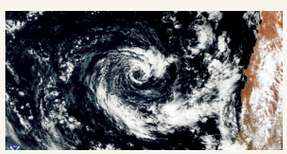
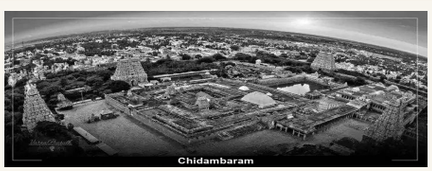
**Aerial photography using drone in the techno world**

### 7.2.2 Satellite Remote Sensing

Satellite remote sensing is the science of collecting data about an object or area from artificial satellites orbiting the Earth. The term 'satellite imagery' refers to digitally transmitted images of the satellites.

### 7.2.3 Global Navigation Satellite System (GNSS)

Have you ever booked a cab using a smart phone app? Did you see the map showing the route of your travel and movement of your vehicle on mobile phones? How is it possible to calculate the time duration of your travel?

| Satellite remote sensing | Aerial Photography |
|---|---|
|  |   |
| ■ High cost of satellite systems. Takes at least 10 years to plan, construct, test and launch.<br>■ Satellites collect large amount of data of the entire area in a short span.<br>■ It allows global coverage and does not require permission.<br>■ Satellites circle the Earth; they can repeat and revisit easily.<br>■ Weather does not affect the functioning of satellites.<br>■ All information is digital; it can be easily integrated with software for image improvement. | ■ Surveying can be planned and executed in a shorter time economically.<br>■ Takes more time to capture an area. Aircraft needs to fly back and forth.<br>■ It covers a small area and needs permission from authorities.<br>■ Revisits or repeatability involves extra cost.<br>■ Adversely affected by bad weather<br>■ It is an analogue record, so no further improvement is possible after obtaining photographs. |



9th_Geography_Unit_7_EM.indd 225 4/4/2022 6:03:38 PM